

NLP-Based System for Legal Analysis

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Introduction

Background: Legal Document Analysis Challenges

- **Manual and time-consuming process** Legal and regulatory documents are long, complex, and require detailed reading.
- **Difficult to scale across countries** Large volumes of documents from different jurisdictions make manual analysis inefficient.
- **Unstructured data formats** Most documents are in PDF format, including scanned files, which are not directly usable.
- **Need for structured policy mapping** RDTII requires linking legal text to specific indicators and policy pillars.

Key Insight

- The main challenge is not analyzing a single document, but processing large collections of legal texts efficiently.
- There is a need for an automated system to convert unstructured legal text into structured, usable data.

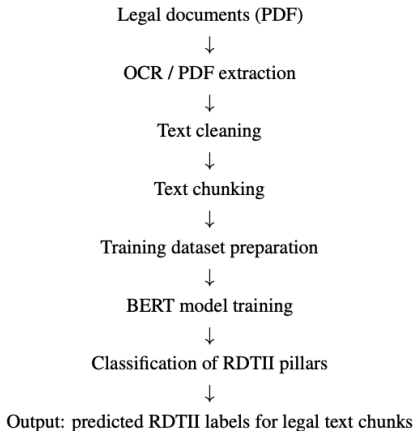
Project Objectives

- **Develop an automated system for legal document analysis** Support the analysis of legal documents within the RDTII framework in a more efficient way.
- **Process and structure legal documents** Convert raw PDF documents into structured text data through extraction and preprocessing.
- **Train classification models** Construct training datasets and train BERT-based models to identify relevant RDTII indicators.
- **Apply the system to new documents** Automatically classify legal text and generate structured outputs for further analysis.

System Overview

System Workflow

The system follows a step-by-step workflow that converts raw legal documents into structured classification results for RDTII analysis.



Key Technologies

- **Natural Language Processing (NLP)** Enables the system to understand and process legal text.
- **Machine Learning** Learns patterns from data to classify legal provisions automatically.
- **BERT-based Models** Transformer-based models designed to capture the context of legal language.
- **Text Preprocessing Pipeline** Includes OCR, text cleaning, and chunking to prepare documents for analysis.
- **Python-based Implementation** Built using standard machine learning libraries for data processing and model training, with PyCharm as the development environment.

Methodology

What is Natural Language Processing (NLP)

- **Understanding human language** NLP enables computers to read, interpret, and process text data.
- **Beyond keyword matching** It captures the meaning and context of text, not just individual words.
- **Applied to legal documents** Helps identify relevant information in long and complex legal texts.
- **Role in this project** Used to understand legal text and support classification into RDTII indicators

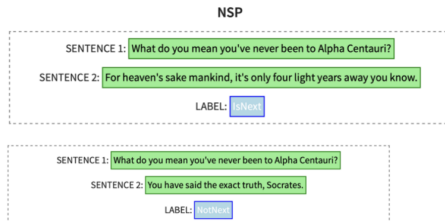
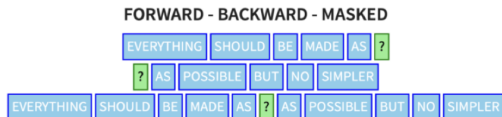
What is BERT and Why We Use It

- **Simple Explanation**

- BERT = Bidirectional Encoder Representations from Transformers
- It reads sentences **contextually**, not just word by word
- Learns meaning from **both sides** of a word

- **Why it is useful for our task**

- Captures legal context and intent, not just matching terms
- Allows automatic classification of long legal documents into RDTII indicators



How We Use BERT in This Project

- **Start from open-source legal model** Use pre-trained Legal-BERT that already understands legal language
- **Adapt to our domain** Incorporate RDTII-related documents and domain resources (guide, keywords)
- **Fine-tune for our task** Train the model to classify legal text into RDTII indicators
- **Apply to new documents** Use the trained model to automatically analyze and classify legal texts

System Implementation

System Implementation Overview

- **Five scripts, five stages** The system is implemented as a modular workflow, where each script handles one processing stage.
- **Stage 1: OCR / PDF reader** Extract text from scanned or digital PDF files.
- **Stage 2: Text cleaning** Standardize extracted text and remove formatting noise.
- **Stage 3: Text chunking** Split documents into structured text sections for analysis.
- **Stage 4: Classification** Apply the trained model to assign RDTII indicators.
- **Stage 5: Output** Export prediction results as structured files for review.

Automation & Flexibility

The system can be run step by step or as a full pipeline. Each stage works independently, but it can also be executed in one click to process documents automatically. This design allows the workflow to scale and efficiently handle large collections of legal texts.

Module 1: OCR / PDF Reader

- Extract text from PDF documents for further processing
- Use OCR for scanned or image-based PDFs; use PDF reader for digital PDFs with selectable text
- Output: structured text for the next stage

Optical Character Recognition (OCR)

- **Goal:** Convert scanned PDF documents — which cannot extract text directly — into machine-readable text for later analysis
- **How it works:**
 - Input:** Scanned legal documents
 - Tool:** Python (using Pytesseract package)
- **Process:**
 1. Convert each scanned page into an image
 2. Apply OCR to detect printed text
 3. Save results as structured .json files initially
 4. Convert freely to other types of files to use

(2) The procuring entity shall, for the purpose of carrying out procurement under the revenue budget of the Government, prepare an annual procurement plan.

(3) Each annual procurement plan prepared under sub-section (1) and (2) shall require the approval of the head of the procuring entity or an officer authorised by him.

(4) The procuring entity shall arrange to publish the procurement plan prepared under this section, following the direction issued by the Government from time to time, for the information of all concerned.

(5) The procuring entity shall not generally split a single procurement work into more than one package, but in order to facilitate a particular procurement activity, it may split a single procurement work into more than one package, and packages into more than one lot.

```
"page": 9,  
"text": "seo cave, UfSe, Gr dv, 2630 8e0e\\n\\n(2) The procuring entity shall, for the ,  
<purpose of carrying out procurement\\nunder the revenue budget of the Government, prepare ,  
<an annual procurement\\nplan. :\\n\\n(3) Each annual procurement plan prepared under ,  
<sub-section (1) and (2)\\nshall require the approval of the head of the procuring entity or,  
<an officer\\nauthorised by him.\\n\\n(4) The procuring entity shall arrange to publish the ,  
<procurement plan\\nprepared under this section, following the direction issued by the ,  
<Government\\nfrom time to time, for the information of all concerned.\\n\\n(5) The ,  
<procuring entity shall not generally split a single procurement work\\ninto more than one ,  
<package, but in order to facilitate a particular procurement\\nactivity, it may split a ,  
<single procurement work into more than one package, and\\ne <\\npackages into more than one ,  
<lot.\\n\\n(6) If a single procurement work is split into more than one package ,
```

Module 2 & 3: Cleaning and Chunking

Module 2: Text Cleaning

- Clean and standardize extracted text
- Remove noise and normalize formatting

Module 3: Text Chunking

- Automatically split text based on legal structure (sections, articles)
- Detect headings and numbering to preserve document logic
- Output: structured text chunks with metadata for model input

Module 4: Classification

Model types

- Single-label model: assign one RDTII indicator per text chunk
- Multi-label model: return multiple possible indicators with confident scores

Application

- Apply trained models to structured text chunks
- Automatically generate predictions for new documents

Scripts

- Dedicated scripts available for standard document prediction
- Also support short-text testing for quick validation

How We Train Our Model

What the model learns

- We teach the model using a **domain dataset (RDTII)**
- Each example shows: **which indicator a legal text belongs to**
- The model learns patterns from these examples

Where the data comes from

- Legal text paired with RDTII labels
- Built from multiple domain resources: RDTII index, guide, keywords, and some tags
- **Main files:**
 - RDTII_2.1_Index_new_041125.xlsx
 - Tags_and_Keywords.xlsx
 - RDTII_Guide_2.1.pdf
 - pillar_dictionary.csv

Output Structure

[illegible]

Single-label output

doc id	pdf pages	section title	test	production 1 score	production 2 score	production 3 score	production 4 score		
rise_invs_v01_part_00001	62	62 Accounts, payments and retention of records	(1) The books of a ship station open for public perusal at all times (2) The books of a ship station and registers of all passengers transported by him. Each passenger shall be identified by a number and date. All passengers of the above origin and of absolute destination, and such further particulars as the Superintendent may require to be shown. (3) The books shall remain closed and undisturbed from unauthorized persons.	7.3	0.9862	6.2	0.94	6.4	0.978
rise_invs_v01_part_00002	65	62 Log to be kept	(1) The books of every travelling station shall, unless exempted by its location, keep a log showing the hours during which the station is in operation. The log shall be accompanied by the names of the station master, the station agent, and the passenger agent.	7.3	0.9148	6.2	0.9098	6.3	0.9032
rise_invs_v01_part_00003	23	3 of the Act	(1) "including any, field, line, garden, enclosure, railway, highway, chabotage, canal, landing, pier, wharf, building, or other premises or place, whether or not intended to be used for or including any harbour, wharf, pier, quay, or other maritime control."	10.4	0.8814	10.1	0.8881	10.2	0.8848
rise_invs_v01_part_00004	80-81	12 Directors for preparation of financial statements	Provisional statement of a company prepared for the purposes of the Act must comply with the following accounting controls: Actual accounting.	5.4	0.8292	5.3	0.8777	11.2	0.8298
rise_invs_v01_part_00005	84	46 Allocation of liability	The Superintendent shall set off as reserve, maintaining the fund limited under these Regulations to be applied by the person to be certified, by the Superintendent.	8.3	0.9146	8.4	0.8881	11.3	0.9183
rise_invs_v01_part_00006	90	13 Powers of Corporation to conduct inquiries	(1) The Corporation shall have the power to conduct inquiries as may be necessary to enforce compliance with these Regulations.	7.5	0.468	6.1	0.382	6.4	0.3882
rise_invs_v01_part_00007	85-86	79 Completed by	(1) The Commission shall have the means of the Corporation under members of the Address Chief General Telephone Fax E-mail	5.3	0.468	5.5	0.2543	12.8	0.3888
rise_invs_v01_part_00008	31	3 Billing advances for Directors	(1) Every Director of the Corporation shall be paid an allowance of \$40 per meeting for every meeting that Director attends. (2) The allowance payable to the Director shall be a charge upon the funds of the Corporation.	5.3	0.4993	5.9	0.3763	12.8	0.3567

Multi-label output

Key Advantages of the System

System Value

Not just a AI model, but a full processing system

- End-to-end workflow from raw document to structured output
- PDF → OCR / extraction → cleaning → chunking → prediction → CSV

Batch processing and scalability

- Automatically processes all files in a folder
- No need to select documents one by one
- Suitable for large-scale document analysis

Automation of key tasks

- OCR for scanned PDFs and extraction for digital files
- Text cleaning, normalization, and structure-based chunking
- Model prediction and structured CSV output

Key Insight

Value comes from automating the full workflow, not only the classification model

Control and Practical Usage

More controlled than general AI

- Predefined RDTII categories (no free-form output)
- Structured predictions with labels and confidence scores
- Measurable, reproducible, and easier to review

Supports human-in-the-loop

- Confidence scores used to rank results
- Analysts focus on the most relevant sections first
- Supports decision-making rather than replacing it

Scalable and adaptable

- Model can be retrained with updated datasets
- Adaptable to new domains and policy frameworks
- Extendable to other projects with similar structure

Core Message

A scalable and controlled system for practical policy analysis

Further Improvements & Conclusion

Further Improvements

- Expand the training dataset to improve coverage and model accuracy
- Strengthen multi-label learning by linking one text to multiple indicators
- Explore more robust models (BERT family and other language models)
- Improve domain filtering and chunking to better capture legal structure
- Extend the system to multilingual legal documents

Conclusion

Final takeaway

This is a scalable system that automates legal document processing and supports policy analysis

- Handles large collections of legal documents efficiently
- Reduces repetitive manual work
- Supports structured analysis and decision-making

Project documents

For more detailed information, please refer to the following project documents:

- Detailed methodology and full pipeline design
`RDIII_Legal_NLP_Project_Documentation.pdf`
- Practical user guide for running the system
`RDIII_Legal_NLP_Classification_User_Guide.pdf`